

Research Notes on Algorithmic bias

Algorithmic bias

>> Section 1

Literature on algorithmic bias has focused on the remedy of fairness, but definitions of fairness are often incompatible with each other and the realities of machine learning optimization. For example, defining fairness as an "equality of outcomes" may simply refer to a system producing the same result for all people, while fairness defined as "equality of treatment" might explicitly consider differences between individuals. As a result, fairness is sometimes described as being in conflict with the accuracy of a model, suggesting innate tensions between the priorities of social welfare and the priorities of the vendors designing these systems. In response to this tension, researchers have suggested more care to the design and use of systems that draw on potentially biased algorithms, with "fairness" defined for specific applications and contexts.

>> Section 2

Algorithms are difficult to define, but may be generally understood as lists of instructions that determine how programs read, collect, process, and analyze data to generate a usable output. For a rigorous technical introduction, see Algorithms. Advances in computer hardware and software have led to an increased capability to process, store and transmit data. This has in turn made the design and adoption of technologies such as machine learning and artificial intelligence technically and commercially feasible. By analyzing and processing data, algorithms are the backbone of search engines, social media websites, recommendation engines, online retail, online advertising, and more. Contemporary social scientists are concerned with algorithmic processes embedded into hardware and software applications because of their political and social impact, and question the underlying assumptions of an algorithm's neutrality. The term algorithmic bias describes systematic and repeatable errors that create unfair outcomes, such as privileging one arbitrary group of users over others. For example, a credit score algorithm may deny a loan without being unfair, if it is consistently weighing relevant financial criteria. If the algorithm recommends loans to one group of users, but denies loans to another set of nearly identical users based on unrelated criteria, and if this behavior can be repeated across multiple occurrences, an algorithm can be described as biased. This bias may be intentional or unintentional (for example, it can come from biased data obtained from a worker that previously did the job the algorithm is going to do from now on).

>> Section 3

Ethics guidelines on AI point to the need for accountability, recommending that steps be taken to improve the interpretability of results. Such solutions include the consideration of the "right to understanding" in machine learning algorithms, and to resist deployment of machine learning in situations where the decisions could not be explained or reviewed. Toward this end, a movement for "Explainable AI" is already underway within organizations such as DARPA, for reasons that go beyond the remedy of bias. Price Waterhouse Coopers, for example, also suggests that monitoring output means designing systems in such a way as to ensure that solitary components of the system can be isolated and shut down if they skew results. An initial approach towards transparency included the open-sourcing of algorithms. Software code can be looked into and improvements can be proposed through source-code-hosting facilities. However, this approach doesn't necessarily produce the intended effects. Companies and organizations can share all possible documentation and code, but this does not establish transparency if the audience doesn't understand the information given. Therefore, the role of an interested critical audience is worth exploring in relation to transparency. Algorithms cannot be held accountable without a critical audience.

>> Section 4

India On July 31, 2018, a draft of the Personal Data Bill was presented. The draft proposes standards for the storage, processing and transmission of data. While it does not use the term algorithm, it makes for provisions for "harm resulting from any processing or any kind of processing undertaken by the fiduciary". It defines "any denial or withdrawal of a service, benefit or good resulting from an evaluative decision about the data principal" or "any discriminatory treatment" as a source of harm that could arise from improper use of data. It also makes special provisions for people of "Intersex status".

>> Section 5

Data bias: Training data may underrepresent certain groups (e.g., minority populations), contain historical inequalities, or be collected in skewed ways, leading the model to perform worse or behave unfairly for those groups. Label bias: Humanprovided labels can encode subjective judgments or prejudices (for example, what is labeled as "risky," "toxic," or "qualified"), so the model learns and amplifies those judgments. Measurement bias: Proxies or measurements used for important concepts (like "creditworthiness" or "job performance") may be noisy or systematically distorted for some groups, which then distorts predictions. Algorithmic bias: Even with relatively balanced data, modeling choices (loss functions,

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thresholds, optimization objectives) can prioritize overall accuracy over fairness, leaving some subgroups with consistently worse outcomes. Deployment bias: A model used outside the context it was designed for (e.g., a model trained on adults applied to children, or one trained in one country deployed in another) can generate biased results because the environment and population differ. Mitigating machine learning bias typically involves interventions at multiple stages: collecting more representative and higher-quality data, auditing datasets and models for disparate error rates or outcomes across groups, adjusting training objectives (such as adding fairness constraints), and monitoring systems after deployment. Transparent documentation of data sources, and intended use cases is also crucial so that users and stakeholders can understand where biases may remain and how to interpret model outputs responsibly.

>> Section 6

Algorithmic bias describes systematic and repeatable harmful tendency in a computerized sociotechnical system to create "unfair" outcomes, such as "privileging" one category over another in ways that may or may not be different from the intended function of the algorithm. Bias can emerge from many factors, including intentionally biased design decisions or the unintended or unanticipated use or decisions relating to the way data is coded, collected, selected or used to train the algorithm. For example, algorithmic bias has been observed in search engine results and social media platforms. This bias can have impacts ranging from privacy violations to reinforcing social biases of race, gender, sexuality, and ethnicity. The study of algorithmic bias is most concerned with algorithms that reflect "systematic and unfair" discrimination. This bias has only recently been addressed in legal frameworks, such as the European Union's General Data Protection Regulation (enforced in 2018) and the Artificial Intelligence Act (proposed in 2021 and adopted in 2024). As algorithms expand their ability to organize society, politics, institutions, and behavior, sociologists have become concerned with the ways in which unanticipated output and manipulation of data can impact the physical world. Because algorithms are often considered to be neutral and unbiased, they can inaccurately project greater authority than human expertise (in part due to the psychological phenomenon of automation bias), and in some cases, reliance on algorithms can displace human responsibility for their outcomes, without last mile thinking. Bias can enter into algorithmic systems as a result of pre-existing cultural, social, or institutional expectations; by how features and labels are chosen; because of technical limitations of their design; or by being used in unanticipated contexts or by audiences who are not considered in the software's initial

design. Algorithmic bias has been cited in cases ranging from election outcomes to the spread of online hate speech. It has also arisen in criminal justice, healthcare, and hiring, compounding existing racial, socioeconomic, and gender biases. The relative inability of facial recognition technology to accurately identify darker-skinned faces has been linked to multiple wrongful arrests of black men, an issue stemming from imbalanced datasets. Problems in understanding, researching, and discovering algorithmic bias persist due to the proprietary nature of algorithms, which are typically treated as trade secrets. Even when full transparency is provided, the complexity of certain algorithms poses a barrier to understanding their functioning. Furthermore, algorithms may change, or respond to input or output in ways that cannot be anticipated or easily reproduced for analysis. In many cases, even within a single website or application, there is no single "algorithm" to examine, but a network of many interrelated programs and data inputs, even between users of the same service. A 2021 survey identified multiple forms of algorithmic bias, including historical, representation, and measurement biases, each of which can contribute to unfair outcomes.